APPENDIX II

CAPE VERDE: DEBT SUSTAINABILITY—UPDATE

1. **Staff has updated Cape Verde's debt sustainability analysis to account for recent economic changes, including the fiscal response to the crisis.** The previous assessment (for the 7th PSI review) found that the risk of debt distress for Cape Verde was low, despite a faster-than-expected increase in debt ratios reducing the comfort level. The updated assessment incorporates the new and lower-than-foreseen end-2009 stock of debt, as well as the new 2010–30 foreign debt service. Macroeconomic assumptions have been revised, including a slightly lower growth rate in 2010, a revised medium-term fiscal framework, and new exchange rate projections in line with the euro depreciation. Moreover, historical nominal GDP numbers have been revised downward, mechanically worsening the debt ratios.

2. **The projected ratios of debt under the baseline and alternative scenarios are similar or slightly more favorable than in the previous DSA, except for the scenario projecting forward the current primary balance.** All debt burden indicators, except the ones associated with the PV of debt-to-GDP, remain well below the thresholds even under the extreme tests. As in the previous DSA, the most extreme shock for the PV of debt-to-GDP—a 30 percent devaluation of the escudo relative to the U.S. dollar—leads to a sustained breach of the threshold. However, this scenario does not incorporate the potential growth and export enhancing effects of the depreciation that would likely offset the direct impact of the higher burden of dollar debt. Also as in the previous DSA, the low growth scenario results in a marginal and temporary breach of the PV of debt-to-GDP threshold. The scenario that fixes the primary balance at the current year's level shows debt ratios that deteriorate relative to the previous DSA given the higher deficit in 2010 relative to 2009. However, staff considers this scenario to be unrealistic given that 2010 represents the peak of the public investment program, and the deficit is projected to decline sharply over the coming years as the major investment projects are completed.

Table 1.: External Debt Sustainability Framework, Baseline Scenario, 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**62.8**	**62.1**	**61.9**			**73.6**	**77.3**	**78.1**	**75.7**	**71.4**	**66.8**		**53.9**	**39.9**	
o/w public and publicly guaranteed (PPG)	46.5	47.7	47.8			62.2	67.0	69.0	67.3	63.8	59.9		47.7	32.6	
Change in external debt	-1.2	-0.7	-0.2			11.7	3.7	0.8	-2.5	-4.3	-4.5		-1.7	-4.8	
Identified net debt-creating flows	-10.4	-9.4	1.1			12.2	7.3	3.1	-0.8	-3.2	-2.7		-0.1	-3.6	
Non-interest current account deficit	**12.6**	**10.6**	**8.5**	**7.8**	**4.0**	**19.4**	**17.8**	**14.7**	**11.6**	**9.0**	**8.2**	**13.4**	**4.8**	**-0.6**	**3.2**
Deficit in balance of goods and services	34.9	30.5	28.6			37.6	35.0	30.6	26.7	23.1	21.6		15.2	8.5	
Exports	42.8	46.2	37.2			37.1	38.7	40.7	42.6	44.4	46.7	41.7	58.0	63.3	**60.2**
Imports	77.8	76.6	65.7			74.7	73.7	71.3	69.3	67.5	68.4	70.8	73.1	71.8	**73.1**
Net current transfers (negative = inflow)	-22.6	-20.7	-22.0	**-24.8**	**2.6**	-20.3	-19.1	-17.4	-16.6	-15.7	-15.0	-17.4	-11.8	-9.3	**-10.7**
o/w official	-5.1	-6.8	-7.6			-6.4	-5.0	-4.5	-3.8	-3.3	-3.1		-2.3	-0.8	
Other current account flows (negative = net inflow)	0.2	0.8	1.9			2.1	1.9	1.5	1.6	1.6	1.6		1.5	0.2	
Net FDI (negative = inflow)	**-14.3**	**-13.6**	**-7.6**	**-8.1**	**3.7**	**-7.0**	**-8.7**	**-9.1**	**-9.2**	**-9.1**	**-8.5**	**-8.6**	**-4.2**	**-2.6**	**-3.1**
Endogenous debt dynamics 2/	-8.7	-6.4	0.2			-0.3	-1.8	-2.5	-3.2	-3.1	-2.4		-0.7	-0.4	
Contribution from nominal interest rate	2.0	2.2	1.5			2.2	2.4	2.4	2.3	2.2	2.1		1.9	1.7	
Contribution from real GDP growth	-4.6	-3.0	-1.8			-2.5	-4.2	-4.9	-5.4	-5.3	-4.4		-2.6	-2.1	
Contribution from price and exchange rate changes	-6.2	-5.5	0.6			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	9.2	8.7	-1.3			-0.5	-3.6	-2.3	-1.7	-1.1	-1.9		-1.6	-1.2	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	47.2			55.7	58.2	58.5	56.9	55.2	52.7		46.0	36.7	
In percent of exports	...	...	126.9			150.3	150.4	143.8	133.5	124.3	112.9		79.3	57.9	
PV of PPG external debt	...	...	33.0			44.3	47.9	49.3	48.5	47.6	45.8		39.8	29.4	
In percent of exports	...	...	88.9			119.6	123.8	121.3	113.8	107.2	98.0		68.6	46.4	
In percent of government revenues	...	...	138.9			190.5	198.8	204.4	203.5	182.9	168.9		146.1	103.8	
Debt service-to-exports ratio (in percent)	8.9	8.5	7.9			10.3	9.6	9.0	7.9	7.7	7.5		7.6	6.9	
PPG debt service-to-exports ratio (in percent)	5.1	4.6	4.2			5.1	4.7	4.5	3.9	4.1	4.2		5.4	5.3	
PPG debt service-to-revenue ratio (in percent)	7.9	7.5	6.6			8.1	7.6	7.7	7.0	7.0	7.3		11.4	11.9	
Total gross financing need (Billions of U.S. dollars)	0.0	0.0	0.1			0.3	0.2	0.2	0.1	0.1	0.1		0.2	0.1	
Non-interest current account deficit that stabilizes debt ratio	13.8	11.3	8.7			7.7	14.1	13.9	14.1	13.2	12.7		6.6	4.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	8.6	5.6	3.0	6.2	2.1	4.1	6.0	6.9	7.6	7.7	6.8	6.5	5.0	5.2	5.0
GDP deflator in US dollar terms (change in percent)	10.6	9.7	-0.9	4.4	10.5	-1.4	0.6	2.2	2.1	2.5	3.0	1.5	3.0	3.1	3.0
Effective interest rate (percent) 5/	3.8	4.0	2.4	4.5	1.2	3.7	3.4	3.4	3.2	3.1	3.2	3.3	3.7	4.2	3.9
Growth of exports of G&S (US dollar terms, in percent)	14.1	24.8	-17.8	17.7	14.2	2.4	11.3	14.9	15.2	15.0	15.7	12.4	11.3	7.2	10.3
Growth of imports of G&S (US dollar terms, in percent)	28.6	14.1	-12.4	12.2	14.9	16.7	5.2	5.7	6.9	7.5	11.3	8.9	7.9	9.0	8.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	27.5	36.2	38.4	40.3	11.4	18.0	28.6	12.9	20.3	15.3
Government revenues (excluding grants, in percent of GDP)	27.6	28.3	23.8			23.3	24.1	24.1	23.8	26.0	27.1		27.2	28.3	27.6
Aid flows (in Billions of US dollars) 7/	0.1	0.1	0.1			0.2	0.2	0.2	0.2	0.1	0.1		0.1	0.1	
o/w Grants	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
o/w Concessional loans	0.1	0.0	0.0			0.1	0.1	0.1	0.1	0.0	0.0		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			10.1	8.9	7.9	6.1	3.7	3.7		2.9	1.4	2.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			50.4	56.1	58.7	64.5	50.9	56.9		41.2	38.1	40.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	1.3	1.5	1.6			1.6	1.7	1.9	2.1	2.3	2.5		3.7	8.1	
Nominal dollar GDP growth	20.2	15.8	2.1			2.7	6.7	9.3	9.9	10.3	10.0	8.1	8.2	8.4	8.1
PV of PPG external debt (in Billions of US dollars)			0.5			0.7	0.8	0.9	1.0	1.1	1.2		1.5	2.4	
(PVt-PVt-1)/GDPt-1 (in percent)						10.0	7.4	6.1	4.0	4.0	2.7	5.7	2.7	0.2	2.0

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2.Cape Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(In percent)

	Projections							
	2010	2011	2012	2013	2014	2015	**2020**	2030
PV of debt-to GDP ratio								
Baseline	44	48	49	48	48	46	**40**	29
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	44	37	32	31	33	34	**31**	18
A2. New public sector loans on less favorable terms in 2010-2030 2	44	51	55	55	53	52	**48**	43
A3. Low growth Scenario 7/	44	49	51	51	51	49	45	36
A4. High Growth Scenario 7/	44	47	48	46	45	43	34	23
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	44	49	52	51	50	48	**42**	31
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	44	50	58	57	54	50	**40**	29
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	44	51	58	57	56	53	**47**	34
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	44	49	50	49	48	46	**40**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	44	48	53	52	51	49	**43**	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	44	68	70	69	68	65	**57**	42
PV of debt-to-exports ratio								
Baseline	120	124	121	114	107	98	**69**	46
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	120	95	80	73	75	74	**53**	29
A2. New public sector loans on less favorable terms in 2010-2030 2	120	132	134	129	120	112	**83**	67
A3. Low growth Scenario 7/	120	125	124	117	111	102	74	51
A4. High Growth Scenario 7/	120	122	119	110	104	94	62	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	120	123	121	114	107	98	**69**	46
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	120	140	172	159	145	128	**82**	55
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	120	123	121	114	107	98	**69**	46
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	120	127	124	116	109	99	**69**	46
B5. Combination of B1-B4 using one-half standard deviation shocks	120	122	126	119	112	102	**72**	49
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	120	123	121	114	107	98	**69**	46
PV of debt-to-revenue ratio								
Baseline	190	199	204	204	183	169	**146**	104
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	190	153	135	131	127	127	**113**	65
A2. New public sector loans on less favorable terms in 2010-2030 2	190	212	227	231	206	193	**177**	150
A3. Low growth Scenario 7/	190	204	212	212	189	176	161	119
A4. High Growth Scenario 7/	190	193	197	195	177	162	130	82
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	190	202	214	213	191	177	**153**	109
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	190	209	242	238	207	185	**147**	104
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	190	212	239	238	214	197	**171**	121
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	190	203	209	207	185	170	**146**	104
B5. Combination of B1-B4 using one-half standard deviation shocks	190	199	218	218	196	181	**157**	112
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	190	282	291	290	260	241	**208**	148

Table 2.Cape Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030 (continued)
(In percent)

Debt service-to-exports ratio

Baseline	5	5	5	4	4	4	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	5	5	4	-1	-3	-3	**4**	3
A2. New public sector loans on less favorable terms in 2010-2030 2	5	5	5	5	4	5	**6**	8
A3. Low growth Scenario 7/	5	5	5	4	4	4	6	6
A4. High Growth Scenario 7/	5	5	4	4	4	4	5	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	5	5	5	4	4	4	**5**	5
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	5	5	6	7	10	9	**7**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	5	5	5	4	4	4	**5**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	5	5	5	4	5	5	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	5	4	4	4	**6**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	5	5	5	4	4	4	**5**	5

Debt service-to-revenue ratio

Baseline	8	8	8	7	7	7	**11**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	8	7	6	-2	-4	-4	**8**	6
A2. New public sector loans on less favorable terms in 2010-2030 2	8	8	8	8	8	9	**13**	17
A3. Low growth Scenario 7/	8	8	8	7	7	8	12	14
A4. High Growth Scenario 7/	8	7	7	7	7	7	10	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	8	8	8	7	7	8	**12**	12
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	8	8	8	10	14	13	**12**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	8	8	9	8	8	9	**13**	14
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	8	8	8	8	8	8	**11**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	8	8	8	7	7	8	**12**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	8	11	11	10	10	10	**16**	17
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	0	0	0	0	0	0	**0**	0

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.
7/ Assumes growth rates respectively 1 percentage points above and below the baseline in 2011-16.

Table 3.Cape Verde: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	68.0	65.6	64.5			80.8	86.6	87.8	84.9	79.7	73.8		58.5	40.5	
o/w foreign-currency denominated	46.5	47.7	47.8			62.2	67.0	69.0	67.3	63.8	59.9		47.7	32.6	
Change in public sector debt	-11.6	-2.5	-1.1			16.3	5.7	1.3	-2.9	-5.2	-5.8		-1.8	-2.9	
Identified debt-creating flows	-12.6	-1.5	-0.8			15.2	4.1	0.1	-3.3	-5.2	-5.7		-2.0	-3.2	
Primary deficit	**-1.0**	**-0.3**	**5.3**	**4.2**	**5.1**	**13.2**	**8.9**	**6.0**	**2.9**	**1.2**	**0.0**	**5.4**	**1.3**	**-1.1**	**0.4**
Revenue and grants	32.8	33.9	29.3	**30.4**	**2.5**	29.6	29.0	28.6	27.7	29.3	30.2	**29.1**	29.5	29.1	**29.4**
of which: grants	5.2	5.6	5.5			6.4	5.0	4.5	3.8	3.3	3.1		2.3	0.8	
Primary (noninterest) expenditure	31.8	33.6	34.6	**34.6**	**3.9**	42.8	38.0	34.6	30.6	30.5	30.1	34.4	30.8	28.1	**29.9**
Automatic debt dynamics	-10.3	-0.5	-6.2			2.1	-4.7	-6.0	-6.3	-6.4	-5.8		-3.3	-2.1	
Contribution from interest rate/growth differential	-6.5	-3.6	-1.7			-2.1	-4.3	-5.5	-6.2	-6.1	-5.1		-2.8	-1.7	
of which: contribution from average real interest rate	-0.2	0.0	0.3			0.5	0.3	0.1	0.1	0.0	-0.1		0.1	0.4	
of which: contribution from real GDP growth	-6.3	-3.6	-1.9			-2.6	-4.6	-5.6	-6.2	-6.0	-5.1		-2.9	-2.2	
Contribution from real exchange rate depreciation	-3.8	3.1	-4.6			4.2	-0.5	-0.4	-0.1	-0.3	-0.6		...	...	
Other identified debt-creating flows	-1.4	-0.7	0.1			-0.1	-0.1	0.0	0.1	0.1	0.1		0.0	0.0	
Privatization receipts (negative)	-1.4	-0.7	0.1			-0.1	-0.1	0.0	0.1	0.1	0.1		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.0	-0.9	-0.3			1.2	1.6	1.2	0.4	-0.1	-0.1		0.2	0.3	
Other Sustainability Indicators															
PV of public sector debt	**21.5**	**17.8**	**49.7**			**62.9**	**67.4**	**68.2**	**66.1**	**63.5**	**59.7**		**50.6**	**37.3**	
o/w foreign-currency denominated	0.0	0.0	33.0			44.3	47.9	49.3	48.5	47.6	45.8		39.8	29.4	
o/w external	...	...	33.0			44.3	47.9	49.3	48.5	47.6	45.8		39.8	29.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.6	4.7	11.2			18.8	14.9	12.0	9.1	8.6	7.2		9.0	7.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	65.6	52.6	169.8			212.5	232.1	238.6	239.0	216.7	198.1		171.5	128.1	
PV of public sector debt-to-revenue ratio (in percent)	78.0	62.9	209.0			270.4	280.0	282.7	277.5	244.0	220.4		185.8	132.0	
o/w external 3/	…	…	138.9			190.5	198.8	204.4	203.5	182.9	168.9		146.1	103.8	
Debt service-to-revenue and grants ratio (in percent) 4/	17.0	14.8	20.2			18.7	20.4	21.2	22.6	25.1	24.0		26.4	30.9	
Debt service-to-revenue ratio (in percent) 4/	20.3	17.7	24.8			23.8	24.6	25.1	26.2	28.3	26.7		28.6	31.8	
Primary deficit that stabilizes the debt-to-GDP ratio	10.6	2.1	6.4	**6.4**	**4.2**	-3.1	3.2	4.7	5.8	6.4	5.8	**3.8**	3.1	1.8	**2.6**
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	8.6	5.6	3.0	6.2	2.1	4.1	6.0	6.9	7.6	7.7	6.8	6.5	5.0	5.2	5.0
Average nominal interest rate on forex debt (in percent)	1.1	1.1	0.0	1.2	0.5	0.8	0.8	1.0	0.9	0.9	0.9	0.9	1.3	2.3	1.5
Average real interest rate on domestic debt (in percent)	2.5	2.3	0.1	4.0	2.8	2.7	4.0	3.1	3.9	3.5	3.5	3.4	3.7	3.5	3.7
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.0	7.0	-9.8	-1.8	12.9	9.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	1.4	2.1	4.7	1.2	2.3	3.4	2.8	2.8	2.7	3.0	3.0	2.9	3.0	3.1	3.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.1	0.1	0.1	0.2	0.3	-0.1	0.0	0.0	0.1	0.1	0.0	0.1	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	27.5	36.2	38.4	40.3	11.4	18.0	28.6	12.9	20.3	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

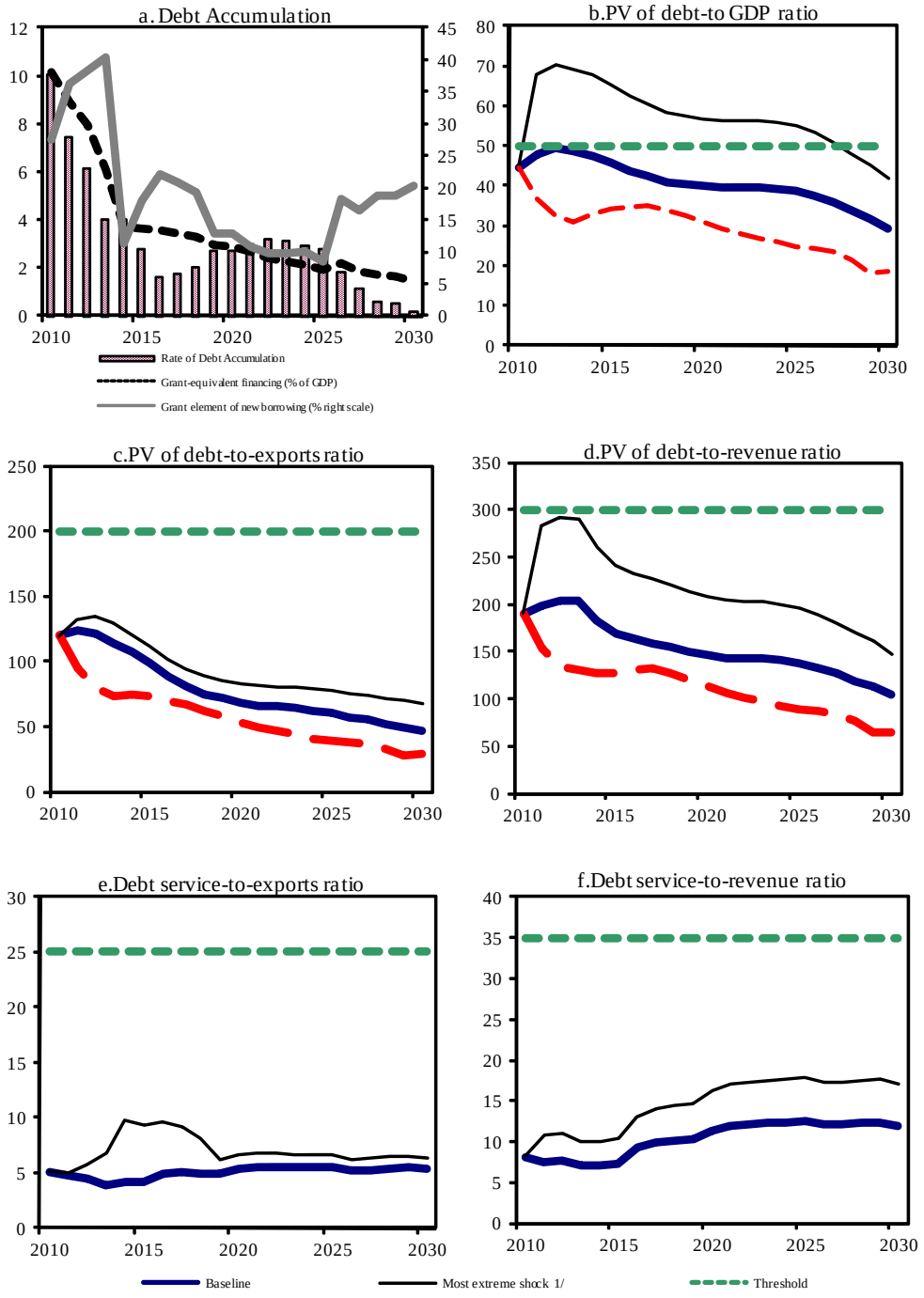
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4.Cape Verde: Sensitivity Analysis for Key Indicators of Public Debt 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	63	67	68	66	63	60	51	37
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	63	63	62	63	65	66	73	88
A2. Primary balance is unchanged from 2010	63	72	81	90	99	108	160	254
A3. Permanently lower GDP growth 1/	63	68	70	69	67	64	62	72
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	63	70	74	73	72	69	67	66
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	63	68	73	71	68	65	57	45
B3. Combination of B1-B2 using one half standard deviation shocks	63	66	69	68	66	63	59	54
B4. One-time 30 percent real depreciation in 2011	63	85	84	81	78	74	65	57
B5. 10 percent of GDP increase in other debt-creating flows in 2011	63	79	79	77	74	70	62	49
PV of Debt-to-Revenue Ratio 2/								
Baseline	212	232	239	239	217	198	172	128
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	212	216	218	228	221	219	247	304
A2. Primary balance is unchanged from 2010	212	249	284	324	338	359	542	872
A3. Permanently lower GDP growth 1/	212	235	244	248	228	213	210	245
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	212	239	256	263	244	229	226	228
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	212	235	255	256	233	215	192	155
B3. Combination of B1-B2 using one half standard deviation shocks	212	227	241	245	226	210	199	186
B4. One-time 30 percent real depreciation in 2011	212	292	293	291	266	245	221	197
B5. 10 percent of GDP increase in other debt-creating flows in 2011	212	270	277	278	253	233	209	170
Debt Service-to-Revenue Ratio 2/								
Baseline	19	20	21	23	25	24	26	31
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	19	20	20	22	21	19	38	62
A2. Primary balance is unchanged from 2010	19	20	22	25	33	39	97	191
A3. Permanently lower GDP growth 1/	19	20	21	23	26	25	31	51
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	19	21	22	24	27	27	35	49
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	19	20	21	23	26	28	30	33
B3. Combination of B1-B2 using one half standard deviation shocks	19	21	21	23	24	23	29	39
B4. One-time 30 percent real depreciation in 2011	19	22	24	25	29	28	37	52
B5. 10 percent of GDP increase in other debt-creating flows in 2011	19	20	23	24	35	34	34	37

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

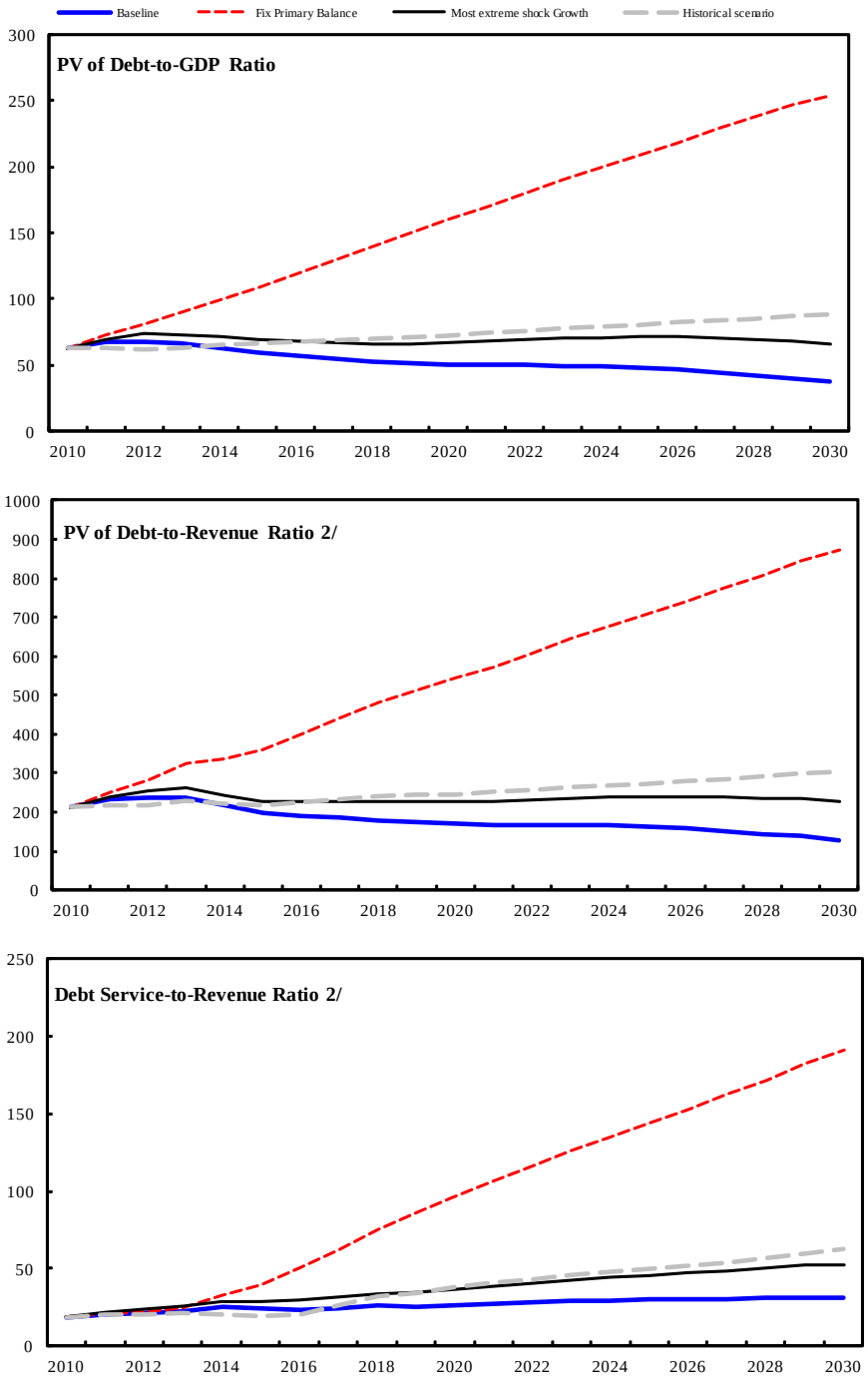
Figure 1. Cape Verde: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

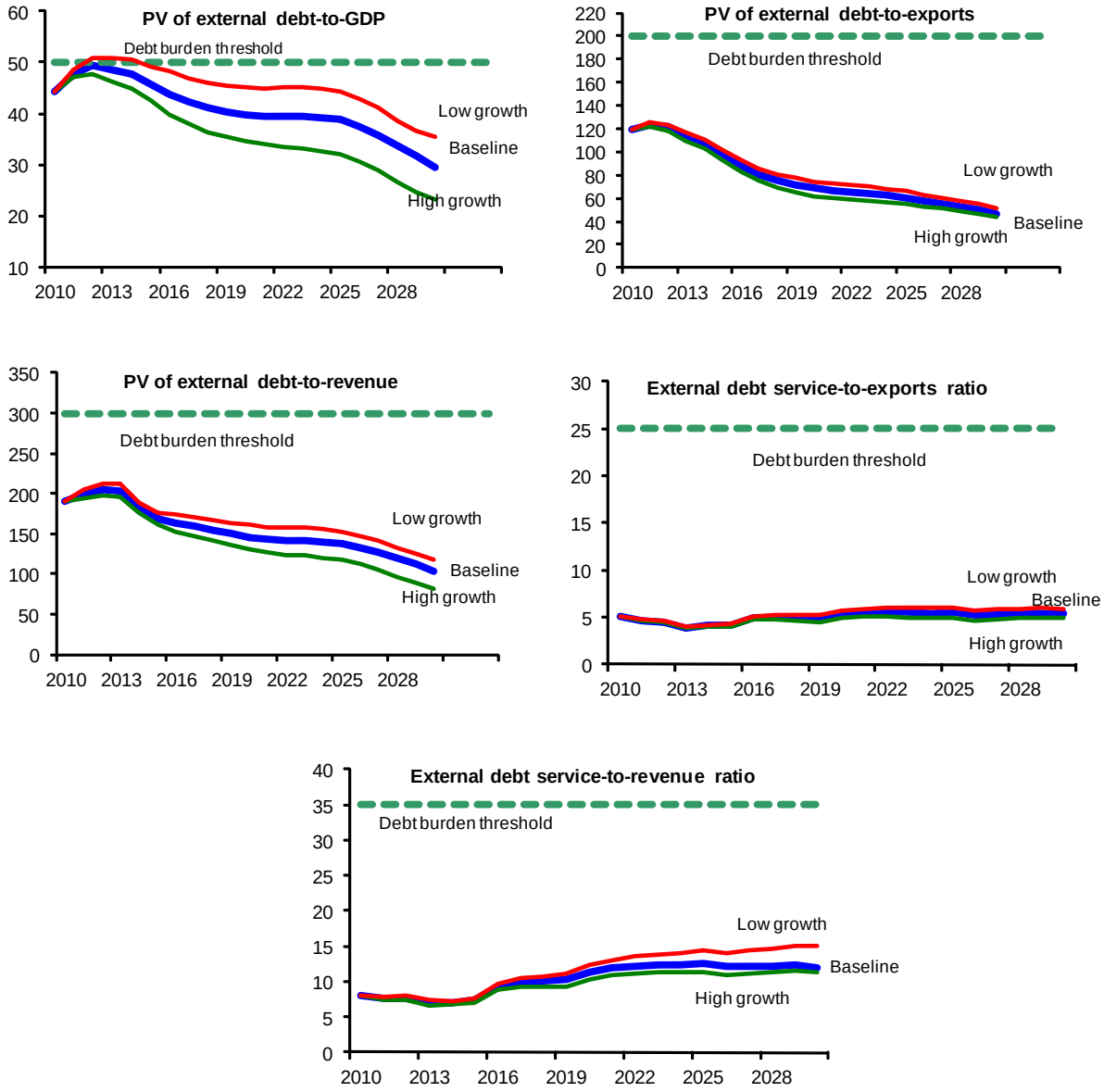
1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2.Cape Verde: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Figure 3. Cape Verde: Debt Sustainability Analysis 1/ 2/



Sources: Cape Verdean authorities and IMF staff estimates.

1/ The DSA does not take into account some remaining amortization due beyond 2030

2/ The high (low) growth scenario assumes a higher (lower) real growth rate by 1 percentage point between 2011 and 2016.